

Kinematics

Def A **lightcone basis** is a basis $\{\bar{n}, n\}$ that satisfies

$$n^2 = \bar{n}^2 = 0 \text{ and } n \cdot \bar{n} = 1$$

The transverse projector $g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n^\nu - \bar{n}^\nu n^\mu$

Def Let $\{n, \bar{n}\}$ be a lightcone basis. Then a 4-vector P can be decomposed as

$$P^\mu = P^+ \bar{n}^\mu + P^- n^\mu + P_T^\mu. \{n, \bar{n}\} \text{ is a } P\text{-collinear basis if}$$

$$[P^+, P^-, \vec{P}_T] \sim Q [1, \lambda^2, \lambda]$$

, where Q is a hard scale and $\lambda = \frac{M}{Q}$

Def For a 4-vector P , a P -collinear basis $\{\bar{n}, n\}$ is a **P -intrinsic basis** if

$$g_T^{\mu\nu} P_\nu = 0$$

i.e. transverse component $P_T^\mu = 0$; here $g_T^{\mu\nu} \equiv g^{\mu\nu} - \bar{n}^\mu n^\nu - \bar{n}^\nu n^\mu$

Prop Given a 4-momentum P , its P -collinear basis is not unique. For any P -collinear basis $\{\bar{n}, n\}$, we can apply the following **Lorentz** transformations to obtain a new P -collinear basis $\{\bar{n}', n'\}$.

Longitudinal rescaling	transverse rotation	tilt $\bar{n}$	tilt n
$\bar{n}' = e^\alpha$ $n = e^{-\alpha}$	$e'_i \rightarrow R_{ij} e_j$	$\bar{n}' = \bar{n} + b_T - \frac{b_T^2}{2} n$	$n = n + b_T - \frac{b_T^2}{2} \bar{n}$
$ \alpha \lesssim 1$		$g_T^{\mu\nu} b_{T\nu} = 0, b_T \sim \lambda$	

Prop Consider we tilt $\bar{n}/n$ by b_T to obtain $\bar{n}'$. The the corresponding transformations are

	tilt $\bar{n}$	tilt n
$\bar{n}'$	$\bar{n}' = \bar{n} + b_T - \frac{b_T^2}{2} n$	$\bar{n}' = \bar{n}$
n'	$n' = n$	$n' = n + b_T - \frac{b_T^2}{2} \bar{n}$
e_i'	$e_i' = e_i - (e_i \cdot b_T) n$	$e_i' = e_i - (e_i \cdot b_T) \bar{n}$

$a^{+'}$	$a^{+'} = a^{+}$	$a^{+'} = a^{+} + a_T \cdot b_T - \frac{b_T^2}{2} a^{-}$
$a^{-'}$	$a^{-'} = a^{-} + a_T \cdot b_T - \frac{b_T^2}{2} a^{+}$	$a^{-'} = a^{-}$
$a_T^{\mu'}$	$a_T^{\mu'} = a_T^{\mu} - a^{+} b_T^{\mu} - b_T \cdot (a_T - a^{+} b_T) n^{\mu}$	$a_T^{\mu'} = a_T^{\mu} - a^{-} b_T^{\mu} - b_T \cdot (a_T - a^{-} b_T) \bar{n}^{\mu}$
$g_T^{\mu\nu'}$	$g_T^{\mu\nu'} = g_T^{\mu\nu} - b_T^{\mu} n^{\nu} - b_T^{\nu} n^{\mu} + b_T^2 n^{\mu} n^{\nu}$	$g_T^{\mu\nu'} = g_T^{\mu\nu} - b_T^{\mu} \bar{n}^{\nu} - b_T^{\nu} \bar{n}^{\mu} + b_T^2 \bar{n}^{\mu} \bar{n}^{\nu}$
$\xi^{\mu\nu'}$	$\xi^{\mu\nu'} = \xi^{\mu\nu} + \xi^{\mu\nu\rho\sigma} b_{T\rho} n_{\sigma}$	$\xi^{\mu\nu'} = \xi^{\mu\nu} - \xi^{\mu\nu\rho\sigma} b_{T\rho} \bar{n}_{\sigma}$

Quark TMDPDF (large P^+)

	P-Intrinsic	Global
$\bar{n}'$		$\bar{n}_P = \bar{n} + \frac{P_T}{P^+} - \frac{P_T^2}{2P^{+2}} n$
e_i'		$e_{Pi} = e_i - \frac{e_i \cdot P_T}{P^+} n$
P	$P = P^+ \bar{n}_P + P_P^- n$	$P = P^+ \bar{n} + P^- n + P_T$
S	$S = S_L \frac{P^+}{M} \bar{n}_P + S_P^- n + S_\perp$	$S = S_L \frac{P^+}{M} \bar{n} + S^- n + \left[\left(S_L \frac{P_T}{M} + S_\perp \right) + \frac{P_T \cdot S_\perp}{P^+} n \right]$
$S_\perp$		$S_\perp = S_T - S_L \frac{P_T}{M} - \frac{1}{P^+} (P_T \cdot S_T - S_L P_T^2) n$
k	$k = x P^+ \bar{n}_P + k_P^- n + k_\perp$	$k = x P^+ \bar{n} + k^- n + \left[(x P_T + k_\perp) + \frac{P_T \cdot k_\perp}{P^+} n \right]$
$k_\perp$		$k_\perp = k_T - x P_T - \frac{1}{P^+} (P_T \cdot k_T - x P_T^2) n$
$g_T'^{\mu\nu}$	$g_T^{\mu\nu} = g^{\mu\nu} - \bar{n}_P^\mu n^\nu - \bar{n}_P^\nu n^\mu$	$g_{TP}^{\mu\nu} = g_T^{\mu\nu} - \frac{P_T^\mu n^\nu + n^\mu P_T^\nu}{P^+} + \frac{P_T^2}{P^{+2}} n^\mu n^\nu$
$\Sigma'^{\mu\nu}$	$\Sigma_P^{\mu\nu} = \Sigma^{\mu\nu\rho\sigma} \bar{n}_{P\rho} n_\sigma$	$\Sigma_P^{\mu\nu} = \Sigma^{\mu\nu} + \Sigma^{\mu\nu\rho\sigma} \frac{P_{T\rho} n_\sigma}{P^+} = \Sigma^{\mu\nu} + \frac{1}{P^+} (\tilde{P}_T^\mu n^\nu - \tilde{P}_T^\nu n^\mu)$

Definition

Given a global P-collinear basis $\{\bar{n}, n\}$, s.p.s

$$P^\mu = [p^+, p^-, p_T^\mu]$$

$$S^\mu = [s_L \frac{p^+}{M}, s^-, s_L \frac{p_T^\mu}{M} + s_\perp^\mu]$$

$$K^\mu = [x p^+, k^-, x p_T^\mu + k_\perp^\mu]$$

, where $k_\perp, s_\perp$ are the intrinsic transverse momentum/spin. We define TMDPDF in the hadron intrinsic frame $\{\bar{n}_p, n\}$, where $\bar{n}_p^\mu = \bar{n}^\mu - \frac{p_T^2}{2p^{+2}} n^\mu + \frac{p_T^\mu}{p^+}$, and

$$p^\mu = p^+ \bar{n}_p^\mu + p^- n^\mu$$

$$s^\mu = s_L \frac{p^+}{M} \bar{n}_p^\mu + s^- n^\mu + s_\perp^\mu$$

$$k^\mu = x p^+ \bar{n}_p^\mu + k^- n^\mu + k_\perp^\mu$$

Def $\Phi_{ab}(x, k_\perp; P, S)$

$$= \int \frac{d\xi^- d\xi_T^2}{(2\pi)^3} e^{i(x p^+ \xi^- - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \langle P, S | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, S \rangle \Big|_{\xi^+ = 0}$$

$$= \frac{1}{2} \left\{ f_1 \bar{n}_p^+ + f_{1T} \frac{\epsilon_{\mu\nu\rho\sigma} \gamma^\mu \bar{n}_p^\nu k_\perp^\rho s_\perp^\sigma}{M} + g_{1S} \gamma^5 \bar{n}_p^+ + h_{1T} i \sigma^{\bar{n}_p s_\perp} \gamma^5 + \frac{h_{1S}^\perp}{M} i \sigma^{\bar{n}_p k_\perp} \gamma^5 + \frac{h_1^\perp}{M} \sigma^{\bar{n}_p k_\perp} \right\}$$

$$= \frac{1}{2} \left\{ \left[f_1 - f_{1T} \frac{\hat{k}_\perp \cdot s_\perp}{M} \right] \bar{n}_p^+ + g_{1S} \gamma^5 \bar{n}_p^+ + \left[h_{1T} s_{\perp\mu} + h_{1S}^\perp \frac{k_{\perp\mu}}{M} + h_1^\perp \frac{\tilde{k}_{\perp\mu}}{M} \right] i \sigma^{\bar{n}_p k_\perp} \gamma^5 \right\}$$

, where

$$g_{1S} = s_L g_{1L} - \frac{k_\perp \cdot s_\perp}{M} g_{1T}, \quad h_{1S}^\perp = s_L h_{1L}^\perp - \frac{k_\perp \cdot s_\perp}{M} h_{1T}^\perp$$

Prop Let $\Phi^\Gamma(z, k_\perp) = \frac{1}{2} \text{Tr}[\Gamma \Phi(z, k_\perp)]$. Then,

Φ^{γ^+}	$\Phi^{\gamma^+ \gamma^5}$	$\Phi^{i \sigma^{\bar{n} k} \gamma^5}$	$\Phi^{i \sigma^{\bar{n} \tilde{k}} \gamma^5}$	$\Phi^{i \sigma^{\bar{n} s} \gamma^5}$
$f_1 - f_{1T} \frac{\hat{k}_\perp \cdot s_\perp}{M}$	g_{1S}	$-(h_{1T} s_\perp \cdot k_\perp + h_{1S}^\perp \frac{k_\perp^2}{M})$	$-(h_{1T} s_\perp \cdot \tilde{k}_\perp + h_1^\perp \frac{\tilde{k}_\perp^2}{M})$	$-h_{1T} s_\perp^2 - \frac{1}{M} (h_{1S}^\perp k_\perp \cdot s_\perp + h_1^\perp \hat{k}_\perp \cdot s_\perp)$

Normalization

Prop At LO partonic level, $f_1 = g_{1L} = h_{1T} = \delta(1-x) \delta^2(\vec{k}_\perp)$, while others are 0.

proof Recall $\int \psi_i(y) |q(p, s)\rangle = u_i(p, s) e^{-iP \cdot y} |0\rangle$
 $\langle q(p, s) | \bar{\psi}_j(y) = \bar{u}_j(p, s) e^{iP \cdot y} \langle 0|$

$$\Phi_{ab}(x, k_\perp; p, s)$$

$$= \int \frac{d\xi^- d\xi_\perp^2}{(2\pi)^3} e^{ik \cdot \xi} \langle p, s | \bar{\psi}_b(0) \psi_a(\xi) | p, s \rangle \Big|_{\xi^+ = 0, k^+ = x p^+}$$

$$= \int \frac{d\xi^- d\xi_\perp^2}{(2\pi)^3} e^{ik \cdot \xi} [u_s(p) \bar{u}_s(p)]_{ab} e^{-iP \cdot \xi}$$

$$= [u_s(k) \bar{u}_s(k)]_{ab} \delta(k^+ - p^+) \delta^2(\vec{k}_\perp - \vec{p}_\perp)$$

$$= \frac{1}{2p^+} (\not{p} + m)(1 + \gamma^5 \not{s}) \delta(1-x) \delta^2(\vec{p}_\perp - \vec{k}_\perp). \text{ Recall } \vec{p}_\perp = 0.$$

$$\textcircled{1} \not{p} = p^+ \not{x}_p + \not{p}_\perp. \text{ At twist 2, } \not{p} = p^+ \not{x}_p$$

$$\frac{1}{2} \text{Tr} [\not{x} \not{p}] = \frac{1}{4p^+} p^+ \text{Tr} [\not{x} \not{x}_p] \delta(1-x) \delta^2(\vec{k}_\perp) = \delta(1-x) \delta^2(\vec{k}_\perp)$$

$$\textcircled{2} m \gamma^5 \not{s} = m \gamma^5 (s^+ \not{x}_p + \not{s}_\perp - s_\perp^i \gamma^i). \text{ At twist 2, } m \gamma^5 \not{s} = s_L p^+ \gamma^5 \not{x}$$

$$\frac{1}{2} \text{Tr} [\not{x} \gamma^5 \not{s}] = \frac{1}{4p^+} s_L p^+ \text{Tr} [\not{x} \gamma^5 \not{x}] \delta(1-x) \delta^2(\vec{k}_\perp) = s_L \delta(1-x) \delta^2(\vec{k}_\perp)$$

$$\textcircled{3} \not{p} \gamma^5 \not{s} = -\not{p} \not{s} \gamma^5 = -p_\mu s_\nu \gamma^\mu \gamma^\nu \gamma^5. \text{ At twist 2,}$$

$$\not{p} \gamma^5 \not{s} = -p_- s_i \not{x}_p \gamma^i \gamma^5 = -p^+ s_i^+ \gamma^i \not{x}_p \gamma^5 = p^+ s_\perp^i i \sigma^{i-} \gamma^5$$

$$\frac{1}{2} \text{Tr} [i \sigma^{i+} \gamma^5 \not{s}] = \frac{1}{4p^+} p^+ s_\perp^i \text{Tr} [i \sigma^{i+} \gamma^5 i \sigma^{i-} \gamma^5] \delta(1-x) \delta^2(\vec{k}_\perp)$$

$$= s_\perp^i \delta(1-x) \delta^2(\vec{k}_\perp)$$

□

Cut - Vertex

$$\begin{aligned}
 \Phi^\Gamma &= \int \frac{d\xi^- d\xi_\perp^2}{2(2\pi)^3} e^{i(xP^+ \xi^- - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \text{Tr}[\Gamma \langle P, s | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, s \rangle |_{\xi^+ = 0}] \\
 &= \int dL^+ d\vec{L}_\perp \delta(xP^+ - L^+) \delta^2(\vec{L}_\perp - \vec{k}_\perp) \\
 &\quad \int \frac{d\xi^- d\xi_\perp^2}{2(2\pi)^3} e^{i(xP^+ \xi^- - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \int \frac{d\ell^+ d\vec{\ell}_\perp}{2\pi} e^{i\ell^+ \xi^-} \text{Tr}[\Gamma \langle P, s | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, s \rangle] \\
 &= \int d^4 L \text{Tr}[\underbrace{\frac{1}{2p^+} \delta(L^+ - x) \delta^2(\vec{L}_\perp - \vec{k}_\perp) \Gamma}_{\text{cut vertex}} \underbrace{\int d^4 \xi e^{i\ell \cdot \xi} \langle P, s | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, s \rangle}_{\hat{T}_{ab}(L; P, s)}]
 \end{aligned}$$

Diagrammatically,

$$\hat{T}_{ab}(L; P, s) = \text{Diagram with two vertices connected by a horizontal line. The left vertex is labeled 'a' and the right vertex is labeled 'b'. A curved line labeled 'L' connects the two vertices above the horizontal line.$$

$$\Phi^\Gamma(x, \vec{k}_\perp; P, s) = \text{Diagram with two vertices connected by a horizontal line. A curved line labeled 'k' connects the two vertices above the horizontal line.}, \text{ where } \int = \frac{1}{2p^+} \frac{\delta(L^+ - x) \delta^2(\vec{L}_\perp - \vec{k}_\perp) \Gamma}{p^+}$$

$$\Phi(x, \vec{k}_\perp; P, s) = \int \frac{d^4 L}{(2\pi)^4} \frac{1}{p^+} \frac{\delta(L^+ - x) \delta^2(\vec{L}_\perp - \vec{k}_\perp)}{p^+} T_\Phi(L; P, s)$$

Quark Collinear PDF (Large P^+)

Def $\Phi_{ab}(x; P, S)$

$$= \int \frac{d\xi^-}{2\pi} e^{i x P^+ \xi^-} \langle P, S | \bar{\psi}_b(0) W(0, \xi^-) \psi_a(\xi^-) | P, S \rangle$$

$$= \frac{1}{2} \left[f_1 \bar{\psi}_p + \not{s}_L g_{1L} \gamma^5 \bar{\psi}_p + h_1 i \not{s}_L \gamma^5 \right]$$

Prop Let $\Phi^\Gamma(x) = \frac{1}{2} \text{Tr} [\Gamma \Phi(x)]$. Then,

Φ^Γ	$\Phi^\Gamma \gamma^5$	$\Phi^\Gamma i \not{s}_L \gamma^5$
f_1	$\not{s}_L g_{1L}$	$-h_1 \not{s}_L^2$

Cut - Vertex

$$\Phi_{ij} = \int \frac{d^4 l}{(2\pi)^4} \underbrace{\delta(l^+ - k^+)}_{\text{cut vertex}} \underbrace{\int d\xi^- e^{i l \cdot \xi} \langle P, S | \bar{\psi}_b(0) W(0, \xi^-) \psi_a(\xi^-) | P, S \rangle}_{\hat{T}(l; P, S)}$$

Diagrammatically,

$$\hat{T}_{ab}(l; P, S) = \text{Diagram: a horizontal line with a vertical cut through it, labeled with } a \text{ and } b \text{ at the ends, and } l \text{ on the cut.}$$

$$\Phi^\Gamma(x; P, S) = \text{Diagram: a horizontal line with a curved cut through it, labeled with } k \text{ on the cut.}, \text{ where } \int = \frac{1}{2} \delta(l^+ - k^+) \Gamma$$

$$\Phi(x; P, S) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{P^+} \delta(l^+ - x P^+) \hat{T}(l; P, S)$$

Quark TMDPDF (large P^-)

	P^- Intrinsic	Global
n'		$n_P = n + \frac{P_T}{P^-} - \frac{P_T^2}{2P^{-2}} \bar{n}$
e_i'		$e_{Pi} = e_i - \frac{e_i \cdot P_T}{P^-} \bar{n}$
P	$P = P^- n_P + P_P^+ \bar{n}$	$P = P^- n + P^+ \bar{n} + P_T$
S	$S = S_L \frac{P^-}{M} n_P + S_P^+ \bar{n} + S_\perp$	$S = S_L \frac{P^-}{M} n + S^+ \bar{n} + \left[(S_L \frac{P_T}{M} + S_\perp) + \frac{P_T \cdot S_\perp}{P^-} \bar{n} \right]$
$S_\perp$		$S_\perp = S_T - S_L \frac{P_T}{M} - \frac{1}{P^-} (P_T \cdot S_T - S_L P_T^2) \bar{n}$
k	$k = x P^- n_P + k_P^+ \bar{n} + k_\perp$	$k = x P^- n + k^+ \bar{n} + \left[(x P_T + k_\perp) + \frac{P_T \cdot k_\perp}{P^-} \bar{n} \right]$
$k_\perp$		$k_\perp = k_T - x P_T - \frac{1}{P^-} (P_T \cdot k_T - x P_T^2) \bar{n}$
$g_T'^{\mu\nu}$	$g_T'^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n_P^\nu - \bar{n}^\nu n_P^\mu$	$g_{TP}^{\mu\nu} = g_T^{\mu\nu} - \frac{P_T^\mu \bar{n}^\nu + \bar{n}^\mu P_T^\nu}{P^-} + \frac{P_T^2}{P^{-2}} \bar{n}^\mu \bar{n}^\nu$
$\Sigma'^{\mu\nu}$	$\Sigma_P^{\mu\nu} = \Sigma^{\mu\nu\rho\sigma} \bar{n}_\rho n_{P\sigma}$	$\Sigma_P^{\mu\nu} = \Sigma^{\mu\nu} - \Sigma^{\mu\nu\rho\sigma} \frac{P_{T\rho} \bar{n}_\sigma}{P^-} = \Sigma^{\mu\nu} + \frac{1}{P^-} (\tilde{P}_T^\mu \bar{n}^\nu - \tilde{P}_T^\nu \bar{n}^\mu)$

Def $\Phi_{ab}(x, k_\perp; P, S)$

$$= \int \frac{d\xi d\xi_T^2}{(2\pi)^3} e^{i(xP^- \xi^+ - \vec{k}_\perp \cdot \vec{\xi}_\perp)} \langle P, S | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, S \rangle \Big|_{\xi^- = 0}$$

$$= \frac{1}{2} \left\{ f_1 n_P - f_{1T} \frac{1}{M} \varepsilon_{\mu\nu\rho\sigma} \bar{n}^\mu n_P^\nu k_\perp^\rho S_\perp^\sigma + g_{1S} \gamma_5 n_P + h_{1T} i \sigma^{\mu\nu} S_{\perp\mu} \gamma_5 + \frac{h_{1S}^\perp}{M} i \sigma^{\mu\nu} k_{\perp\mu} \gamma_5 - \frac{h_{1T}^\perp}{M} i \sigma^{\mu\nu} k_{\perp\mu} n_P^\nu \right\}$$

$$= \frac{1}{2} \left\{ [f_1 - f_{1T} \frac{\vec{k}_\perp \cdot S_\perp}{M}] n_P + g_{1S} \gamma_5 n_P + [h_{1T} S_{\perp\mu} + h_{1S}^\perp \frac{k_{\perp\mu}}{M} + h_{1T}^\perp \frac{\tilde{k}_{\perp\mu}}{M}] i \sigma^{\mu\nu} \gamma_5 \right\}$$

, where

$$g_{1S} = S_L g_{1L} - \frac{k_\perp \cdot S_\perp}{M} g_{1T}, \quad h_{1S}^\perp = S_L h_{1L}^\perp - \frac{k_\perp \cdot S_\perp}{M} h_{1T}^\perp$$

Prop

$\Phi^{\bar{\pi}}$	$\Phi^{\bar{\pi}\gamma^5}$	$\Phi^{i\bar{6}\bar{n}k\gamma^5}$	$\Phi^{i\bar{6}\bar{n}\tilde{k}\gamma^5}$	$\Phi^{i\bar{6}\bar{n}s\gamma^5}$
$\frac{f_1 - f_T^\perp \hat{k}_\perp \cdot s_\perp}{M}$	g_{1s}	$-(h_{1T}s_\perp \cdot k_\perp + h_{1s}^\perp k_\perp^2)$	$-(h_{1T}s_\perp \cdot \hat{k}_\perp + h_{1i}^\perp \hat{k}_\perp^2)$	$-h_{1T}s_\perp^2 - \frac{1}{M} (h_{1s}^\perp k_\perp \cdot s_\perp + h_{1i}^\perp \hat{k}_\perp \cdot s_\perp)$

Cut - Vertex

$$\Phi = \int \frac{d^4 l}{(2\pi)^4} \underbrace{\delta(\bar{l} - \bar{k}^-) \delta^2(\bar{l}_\perp - \bar{k}_\perp)}_{\text{cut vertex}} \underbrace{\int d^4 \xi e^{i l \cdot \xi} \langle P, s | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, s \rangle}_{\hat{T}(k; P, s)}$$

Diagrammatically,

$$\hat{T}_{ab}(l; P, s) = \text{Diagram: two vertices connected by a horizontal line with a vertical line crossing it, labeled with } a, l, b \text{ at the ends.}$$

$$\Phi^\Gamma(x, k_\perp; P, s) = \text{Diagram: similar to the previous one but with a curved line labeled } k \text{ above the horizontal line,} \text{ where } \int = \frac{1}{2} \delta(\bar{l} - \bar{k}^-) \delta^2(\bar{l}_\perp - \bar{k}_\perp) \Gamma$$

$$\Phi(x, \vec{k}_\perp; P, s) = \int \frac{d^4 l}{(2\pi)^4} \frac{1}{P^-} \delta(\bar{l}^- - x) \delta^2(\bar{l}_\perp - \vec{k}_\perp) \hat{T}_\Phi(l; P, s)$$

Quark Collinear PDF (large P^-)

Def $\Phi_{ab}(x; P, S)$

$$= \frac{\int d\xi^+}{2\pi} e^{i x P^- \xi^+} \langle P, S | \bar{\psi}_b(0) W(0, \xi^+) \psi_a(\xi^+) | P, S \rangle$$

$$= \frac{1}{2} [f_1 \not{P} + \not{S}_\perp g_{1\perp} \gamma^5 \not{P} + h_1 i \not{S}_\perp \gamma^5]$$

Prop Let $\Phi^\Gamma(x) = \frac{1}{2} T_\Gamma [\Gamma \Phi(x)]$. Then,

$\Phi^{\bar{\Gamma}}$	$\Phi^{\bar{\Gamma} \gamma^5}$	$\Phi^{i \bar{\Gamma} \gamma^5}$
f_1	$S_\perp g_{1\perp}$	$-H_1 S_\perp^2$

Cut - Vertex

Φ_{ij}

$$= \frac{\int d^4 l}{(2\pi)^4} \underbrace{\delta(l^- - k^-)}_{\text{cut vertex}} \underbrace{\int d^4 \xi e^{i l \cdot \xi} \langle P, S | \bar{\psi}_b(0) W(0, \xi) \psi_a(\xi) | P, S \rangle}_{\hat{T}(k; P, S)}$$

Diagrammatically,

$\hat{T}_{ab}(l; P, S) =$

$\Phi^\Gamma(x; P, S) =$, where $\int = \frac{1}{2} \delta(l^- - k^-) \Gamma$

$$\Phi(x; P, S) = \int d^4 l \frac{1}{(2\pi)^4} \frac{\delta(l^- - x P^-)}{P^-} T_\Phi(l; P, S)$$

$$\frac{\int d^4 l}{(2\pi)^4} T_\Phi(l; P, S) = \int dx P^- \Phi(x; P, S)$$

Quark TMDFF (large k^-)

	k -Intrinsic	P -Intrinsic
n'		$n_k = n_p + \frac{k_\perp}{k^-} - \frac{k_\perp^2}{2k^{-2}} \bar{n}$
e_i'		$e_{ki} = e_{pi} - \frac{e_{pi} \cdot k_\perp}{k^-} \bar{n}$
k	$k = k^- n_k + k_k^+ \bar{n}$	$k = k^- n_p + k_p^+ \bar{n} + k_\perp$
S	$S = S_L \frac{z k^-}{M} n_k + S_k^+ \bar{n} + S_{\perp k}$	$S = S_L \frac{z k^-}{M} n_p + S_p^+ \bar{n} + \left[(S_L \frac{z k_\perp}{M} + S_{\perp k}) + \frac{k_\perp \cdot S_{\perp k} \bar{n}}{k^-} \right]$
$S_\perp$		$S_{\perp k} = S_\perp - S_L \frac{z k_\perp}{M} - \frac{1}{k^-} (k_\perp \cdot S_\perp - S_L \frac{k_\perp^2}{2M}) \bar{n}$
P	$P = z k^- n_k + P_k^+ \bar{n} + P_{\perp k}$	$P = z k^- n_p + P_p^+ \bar{n}$
$P_\perp$		$P_{\perp k} = -z k_\perp + z \frac{k_\perp^2}{k^-} \bar{n}$
$g_T^{\mu\nu}$	$g_{TK}^{\mu\nu} = g^{\mu\nu} - \bar{n}^\mu n_k^\nu - \bar{n}^\nu n_k^\mu$	$g_{TP}^{\mu\nu} = g^{\mu\nu} - \frac{k_\perp^\mu \bar{n}^\nu + \bar{n}^\mu k_\perp^\nu}{k^-} + \frac{k_\perp^2}{k^{-2}} \bar{n}^\mu \bar{n}^\nu$
$\Sigma^{\mu\nu}$	$\Sigma_K^{\mu\nu} = \Sigma^{\mu\nu\rho\sigma} \bar{n}_\rho n_{k\sigma}$	$\Sigma_P^{\mu\nu} = \Sigma^{\mu\nu} - \Sigma^{\mu\nu\rho\sigma} \frac{k_{\perp\rho} \bar{n}_\sigma}{k^-} = \Sigma^{\mu\nu} + \frac{1}{k^-} (\tilde{k}_\perp^{\mu-\nu} - \tilde{k}_\perp^{\nu-\mu})$

Note that we can define $P_\perp^\mu = g_{TP}^{\mu\nu} P_{\perp k\nu} = -z k_\perp^\mu$. Thus,

$$P_\perp = -z k_\perp$$

Definition

Consider in P-intrinsic basis $\{\bar{n}, n_P\}$

$$P = z k^- n_P + P_P^+ \bar{n}$$

$$k = k^- n_P + k_P^+ \bar{n} + k_\perp$$

$$s = \frac{s_\perp}{z} \frac{k}{M} n_P + s_P^+ \bar{n} + s_\perp$$

Def $\Delta_{ij}(z, P_\perp = -z k_\perp; P, S)$

$$= \frac{1}{2z} \int dP S_X \int \frac{dy^+ dy_\perp}{(2\pi)^3} e^{i(k^- y^+ - \bar{k}_\perp \cdot \bar{y}_\perp)} \langle 0 | W(0, y) \psi_i(y) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) W(0, \infty) | 0 \rangle |_{y^- = 0}$$

$$= \frac{1}{2} \left\{ D_1 n_P - D_{1T}^\perp \frac{\epsilon_{\mu\nu\rho\sigma} \gamma^\mu n_P^\nu P_\perp^\rho s_\perp^\sigma}{M} + G_{1S} \gamma^5 n_P + H_{1T} i \sigma^{\mu\nu} s_\perp \gamma^5 + \frac{H_{1S}^\perp}{M} i \sigma^{\mu\nu} P_\perp \gamma^5 - \frac{H_1^\perp}{M} \sigma^{\mu\nu} P_\perp n_P \right\}$$

$$= \frac{1}{2} \left\{ [D_1 + D_{1T}^\perp \frac{z \hat{k}_\perp \cdot s_\perp}{M}] n_P + G_{1S} \gamma^5 n_P + [H_{1T} s_{\perp\mu} - \frac{z}{M} (H_{1S}^\perp k_{\perp\mu} + H_1^\perp \hat{k}_\perp)] i \sigma^{\mu\nu} \gamma^5 \right\}$$

, where

$$G_{1S} = s_\perp G_{1L} + z \frac{k_\perp \cdot s_\perp}{M} G_{1T}, \quad H_{1S}^\perp = s_\perp H_{1L}^\perp + z \frac{k_\perp \cdot s_\perp}{M} H_{1T}^\perp$$

Prop Let $\Delta^\Gamma(z, k_\perp) = \frac{1}{z} T_V[\Gamma \Delta(z, k_\perp)]$. Then,

$\Phi^{\bar{n}}$	$\Phi^{\bar{n} \gamma^5}$	$\Phi^{i \sigma^{\mu\nu} \hat{k}_\perp \gamma^5}$	$\Phi^{i \sigma^{\mu\nu} \hat{k}_\perp \gamma^5}$	$\Phi^{i \sigma^{\mu\nu} \hat{k}_\perp \gamma^5}$
$D_1 + D_{1T}^\perp \frac{z \hat{k}_\perp \cdot s_\perp}{M}$	G_{1S}	$-H_{1T} s_\perp \cdot \hat{k}_\perp + \frac{z}{M} H_{1S}^\perp k_\perp^2$	$-H_{1T} s_\perp \cdot \hat{k}_\perp + \frac{z}{M} H_{1S}^\perp \hat{k}_\perp^2$	$-H_{1T} s_\perp^2 + \frac{z}{M} (H_{1S}^\perp k_\perp \cdot s_\perp + H_1^\perp \hat{k}_\perp \cdot s_\perp)$

Normalization

Prop At LO partonic level, $D_1 = G_{1L} = H_{1T} = \frac{1}{2} \delta(1-z) \delta^2(\vec{k}_\perp)$, while others are 0.

The different normalization is because we average over initial spins, but sum over final.

proof Recall $\int \psi_i(y) |q(P, s)\rangle = u_i(P, s) e^{-iP \cdot y} |0\rangle$
 $\langle q(P, s) | \bar{\psi}_j(y) = \bar{u}_j(P, s) e^{iP \cdot y} \langle 0|$

$$\Delta_{ij}(x, k_\perp; P, S)$$

$$= \frac{1}{2z} \int dP^+ \int d^2y_\perp \frac{e^{i(k^- y^+ - \vec{k}_\perp \cdot \vec{y}_\perp)}}{(2\pi)^3} \langle 0 | \psi_i(y) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) | 0 \rangle |_{y^- = 0}$$

$$= \frac{1}{2z} \int d^2y_\perp \frac{e^{i(k^- y^+ - \vec{k}_\perp \cdot \vec{y}_\perp)}}{(2\pi)^3} [u_s(P) \bar{u}_s(P)]_{ij} e^{-iP \cdot y^+} = \frac{1}{2z} [u_s(k) \bar{u}_s(k)]_{ij} \delta(k^- - P^-) \delta^2(\vec{k}_\perp)$$

$$= \frac{1}{4P^-} (\not{P} + m) (1 + \gamma^5 \not{S}) \delta(1-z) \delta^2(\vec{k}_\perp)$$

This is exactly half of the TMDPDF case

□

Cut - Vertex

$$\Delta_{ij} = \int d^4l \frac{1}{(2\pi)^4} \underbrace{\delta(l^- - k^-) \delta^2(\vec{l}_\perp - \vec{k}_\perp)}_{\text{cut vertex}} \underbrace{\int dP^+ \int d^2y_\perp e^{iL \cdot y} \langle 0 | W(w, y) \psi_i(y) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) W(0, w) | 0 \rangle}_{\hat{T}_{ij}(L; P, S)}$$

Diagrammatically,

$$\hat{T}_{ab}(L; P, S) =$$

$$\Phi^F(x, k_\perp; P, S) =$$

, where $\int = \frac{1}{4z} \delta(l^- - k^-) \delta^2(\vec{l}_\perp - \vec{k}_\perp) \Gamma$

$$\Delta(z, k_T; P_h, S_h) = \int d^4l \frac{1}{(2\pi)^4} \frac{1}{2z} \delta(l^- - P_h^-) \delta^2(\vec{l}_\perp - \vec{k}_\perp) T_\Delta(l; P, S)$$

Quark Collinear FF (large k^-)

Def $\Delta_{ij}(z; P, S)$

$$= \frac{z}{2} \int dP_+ x \int \frac{dy^+}{2\pi} e^{ik^- y^+} \langle 0 | W(\infty, y^+) \psi_i(y^+) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) W(0, \infty) | 0 \rangle$$

$$= \frac{1}{2} [D_1 \not{P}_+ \not{S}_\perp \not{G}_{1\perp} \gamma^5 \not{P}_+ + H_1 i \not{S}_\perp \gamma^5]$$

Prop Let $\Delta^\Gamma(z) = \frac{1}{z} T_V [\Gamma \Delta(z)]$. Then,

$\bar{\psi}$	$\bar{\psi} \gamma^5$	$\bar{\psi} i \not{S}_\perp \gamma^5$
D_1	$\not{S}_\perp \not{G}_{1\perp}$	$-H_1 \not{S}_\perp^2$

Cut - Vertex

$$\Delta_{ij} = \int \frac{d^4 l}{(2\pi)^4} \underbrace{\frac{z}{2} \delta(l^- - k^-)}_{\text{cut vertex}} \underbrace{\int dP_+ x \int \frac{dy^+}{2\pi} e^{il \cdot y} \langle 0 | W(\infty, y) \psi_i(y) | P, S; X \rangle \langle P, S; X | \bar{\psi}_j(0) W(0, \infty) | 0 \rangle}_{\hat{T}_{ij}(l; P, S)}$$

Diagrammatically,

$$\hat{T}_{ab}(l; P, S) = \text{Diagram: A circle with a vertical line through its center. The top of the vertical line has an upward arrow labeled 'P, S'. The bottom of the vertical line has a downward arrow labeled 'j'. The left side of the circle has an incoming arrow labeled 'j'. The right side of the circle has an outgoing arrow labeled 'i'.$$

$$\Delta^\Gamma(x, k_\perp; P, S) = \text{Diagram: A circle with a vertical line through its center. The top of the vertical line has an upward arrow labeled 'P, S'. The bottom of the vertical line has a downward arrow labeled 'j'. The left side of the circle has an incoming arrow labeled 'j'. The right side of the circle has an outgoing arrow labeled 'i'. A curved arrow on the bottom of the circle indicates a cut.}, \text{ where } \int = \frac{z}{4} \delta(l^- - k^-) \Gamma$$

$$\Delta_{ij}(z; P, S) = \int \frac{d^4 l}{(2\pi)^4} \frac{z}{2} \delta(l^- - \frac{P^-}{z}) \hat{T}_{ij}(l; P, S) = \int \frac{d^4 l}{(2\pi)^4} \frac{z^3}{2P^-} \delta(z^- - \frac{P^-}{l^-}) \hat{T}_{ij}(l; P, S)$$

$$\int \frac{d^4 l}{(2\pi)^4} \hat{T}_{ij}(l; P, S) = \int dz \frac{2P^-}{z^3} \Delta_{ij}(z; P, S)$$

